BRIEFING NOTE

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Income Support Caseload Forecast — Key Findings, Model Confidence, and Interpretation Guidance

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RE: Stage 1 Forecasting System — 24-Month Outlook for Income Support Caseload

PURPOSE

This note communicates the principal findings of the automated Income Support caseload forecasting system (`caseload-forecast-demo`, Stage 1). It answers three questions for decision-makers: what does the dominant model predict over the next two years, how reliably should those predictions be treated, and what contextual factors must be kept in mind when interpreting the numbers? A full technical appendix (`note-1-appendix.html`) substantiates every claim made here with charts, diagnostic tables, and data provenance details.

KEY FINDINGS

The ARIMA model projects the Income Support caseload will remain **essentially stable at approximately 61,000 cases** through September 2027. The September 2025 baseline of 61,061 cases is expected to change by fewer than 200 cases (+0.2%) at the 24-month central estimate. This is not a forecast of growth or decline — it is a forecast of *stability* under current conditions.

Horizon	Month	Forecast	95% Pred. Interval	vs Now	% Change
Current	Sep 2025	61,061	—	—	—
+6 months	Mar 2026	61,794	54,164 – 70,499	+733	+1.2%
+12 months	Sep 2026	61,237	48,251 – 77,719	+176	+0.3%
+18 months	Mar 2027	61,497	42,700 – 88,570	+436	+0.7%
+24 months	Sep 2027	61,211	37,831 – 99,039	+150	+0.2%

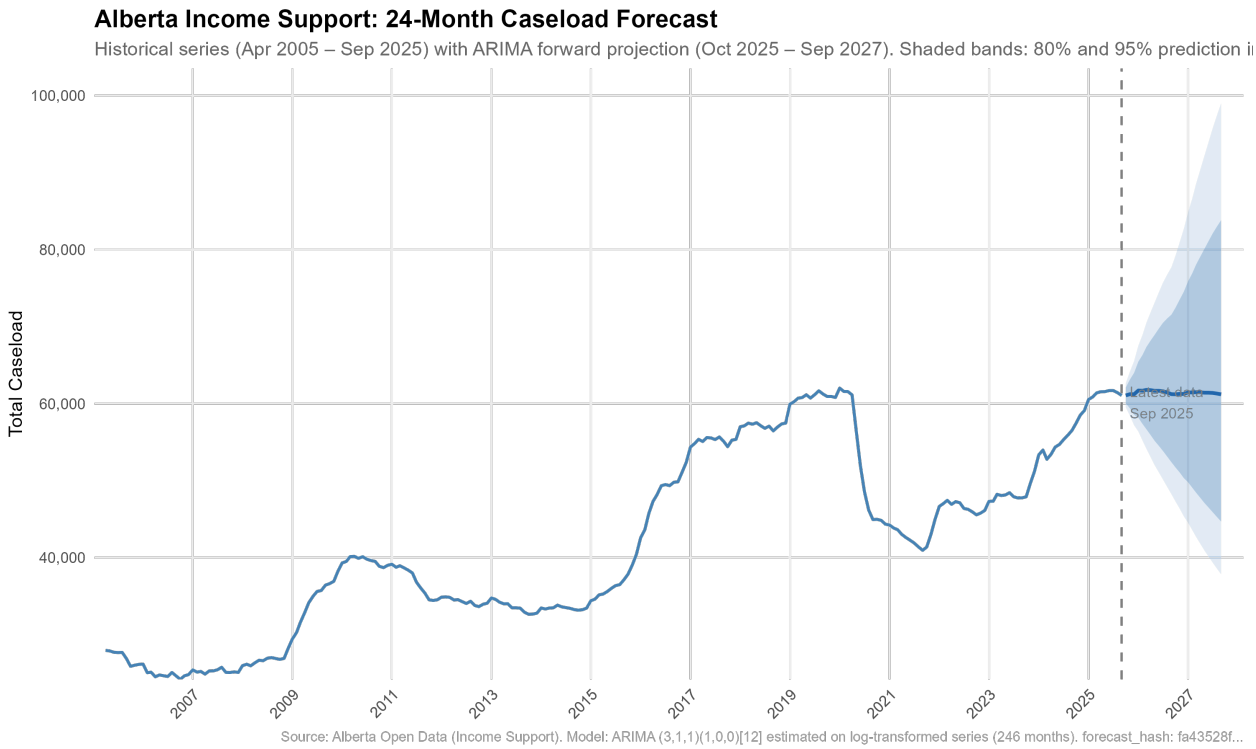

Figure 1. Alberta Income Support: 20-year history (teal) with 24-month ARIMA projection. Shaded bands show 80% and 95% prediction intervals. Dashed line marks September 2025 (latest data).

The 95% prediction interval widens substantially with time: approximately $±$8,200 at six months, growing to $±$30,000 at 24 months. This expanding range is expected — it reflects the genuine uncertainty inherited from 20 years of historical volatility — and should inform how management uses these numbers (see Recommended Actions).

MODEL CONFIDENCE

Dominant model: ARIMA(3,1,1)(1,0,0)[12]. Two competing models were evaluated. The *Seasonal Naive* baseline (Tier 1) repeats the previous year's seasonal pattern without trend. The *ARIMA* model (Tier 2) was automatically selected from the full 20-year history (222 months of training data) on a log-transformed series, capturing both the trend and the annual seasonal cycle. It is the recommended model for planning purposes.

Model	Specification	RMSE	MAE	MAPE
Tier 1: Naive Baseline	Seasonal naive (repeat last year)	10,300	9,545	16.4%
Tier 2: ARIMA	(3,1,1)(1,0,0)[12] on log series	8,639	7,831	13.3%

{*Metrics from 24-month backtest validation (October 2023 – September 2025, held out during training). RMSE = root mean squared error; MAE = mean absolute error; MAPE = mean absolute percentage error.*}

The ARIMA model reduces MAPE by three percentage points relative to the Naive baseline, translating to

roughly 1,700 fewer cases of average monthly error. That improvement is meaningful for budget planning, though the 13.3% MAPE (approximately 8,100 cases per month on average) also signals that precision is limited: **these are directional forecasts, not exact counts.**

INTERPRETATION GUIDANCE

Four factors should shape how leadership reads the forecast table:

1. Stability is the central story, not a default. The ARIMA model's flat trajectory reflects the recent trend: caseload stabilised in the 60,000-62,000 range following the post-COVID normalisation in 2022. The forecast extends that observed stability forward. It is not an artifact of model conservatism.

2. Prediction intervals reflect real-world volatility. The 95% prediction interval at 24 months spans nearly 61,000 cases (from 37,831 to 99,039). These wide bounds are not a modelling failure. They honestly represent a system that has experienced a +50% change during COVID-19 and a $-$20% change during post-pandemic normalisation. If history repeats at a similar magnitude, the actual caseload in 2027 could land anywhere in that range.

3. Short horizons are substantially more reliable. At six months, the 95% interval spans ~16,000 cases (±13% of current). At 24 months it spans ~61,000 cases (±50%). Six- to twelve-month forecasts are appropriate for operational planning; twenty-four month forecasts are best used for scenario bookending.

4. The forecast is not policy-sensitive. This model captures statistical patterns in historical data. It cannot anticipate a new structural break — an economic recession, an oil price shock, a pandemic, or a program policy change. When such events occur, the model must be retrained with updated data to recalibrate.

RECOMMENDED ACTIONS

1. **Use central forecast for baseline planning; use the 95% upper bound as the contingency ceiling.** At +12 months the upper bound is 77,719 cases — a reasonable stress-test scenario given historical precedent.
2. **Schedule quarterly model refresh.** The pipeline is fully automated and can regenerate all forecasts from updated open data with a single command (`Rscript flow.R`). A quarterly cycle ensures the forecast incorporates the most recent trend.
3. **Commission Tier 3 (client-type disaggregation) for program planning.** Income Support comprises four distinct client populations with different drivers. Forecasting each separately (and summing back to the total) would provide program managers with caseload trajectories tailored to their service lines, and is the planned next model tier.

AI Assistance Disclosure: This document was drafted with the assistance of an AI coding assistant (GitHub Copilot, Claude) for structure, language refinement, and alignment with project documentation. All content reflects the author's professional judgment and was reviewed for accuracy against pipeline output artifacts.